PROJECT REPORT · ANALYTICS CASE STUDY

OTT CUSTOMER CHURN ANALYSIS

Customer Retention & Risk Analytics

Python | Power BI | DAX | SQL / SQLite | EDA

PREPARED BY

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01 · INTRODUCTION

PROJECT OVERVIEW

This project analyzes customer, subscription and support data from an OTT streaming platform to understand customer churn, identify high-risk segments, investigate cancellation reasons, measure revenue at risk and generate actionable retention recommendations.

Python was used for data preparation, feature engineering, exploratory analysis and visualization, while Power BI was used for data modeling, DAX measures and the interactive dashboard. Together they form a complete workflow from raw data to business insight.

▸ What I Built

An end-to-end churn analysis: cleaned and engineered a 3-table dataset in Python, explored churn patterns across plans, contracts, geography and support activity, defined core KPIs, and delivered an interactive Power BI dashboard for exploring the results.

▸ Why It Matters

OTT revenue is recurring. Every churned customer takes a stream of future monthly charges with them, so understanding *who* churns, *why* they leave and *how much revenue* is exposed is the first step toward protecting it.

▸ Project at a Glance

CUSTOMERS	SUBSCRIPTIONS	SUPPORT RECORDS	CONNECTED TABLES	TOOLCHAIN
10,000	10,000	6,494	3	Python + PBI

How to read this report: it follows the actual project flow — problem → data → analysis → dashboard → findings → recommendations. All figures come from the project notebook and dashboard; no external industry statistics were used.

02 · FRAMING

PROBLEM STATEMENT & OBJECTIVES

▸ Problem Statement

Customer churn directly affects recurring revenue in subscription-based OTT platforms. The goal of this project is to understand **which customers are churning**, **why they are leaving**, **which segments are at greater risk** and **how much revenue is exposed**.

Objectives

1. Measure overall churn and retention.
2. Identify high-risk customer segments.
3. Analyze cancellation reasons.
4. Examine support activity and its relationship with churn.
5. Analyze churn across plans, contracts, gender and states.
6. Analyze monthly churn patterns.
7. Quantify revenue at risk.
8. Develop practical retention recommendations.

▸ Scope & Approach

The analysis is descriptive and diagnostic: it uses group comparisons, trends and correlation to surface churn patterns. No predictive machine-learning model was built in this project — churn risk is derived from the dataset's churn score, and all findings are reported as observed associations, not proven causes.

03 · QUESTIONS

BUSINESS QUESTIONS

Eight questions guided the analysis. Each one maps directly to a section of the notebook or a visual on the dashboard.

1. What percentage of customers are churning?
2. Which customer segments have higher churn?
3. Why are customers cancelling?
4. Is support escalation associated with higher churn?
5. How does churn vary across states?
6. How does churn change over time?
7. How much revenue is associated with churned customers?
8. What actions could improve customer retention?

Evidence standard: every answer in this report is grounded in the project data — computed KPIs, group-level churn rates and dashboard visuals. Where the data shows a relationship rather than a cause (e.g., escalations and churn), it is described exactly that way.

04 · DATA

DATASET & DATA MODEL

DB_CUSTOMER
10,000 rows
Demographics & profile

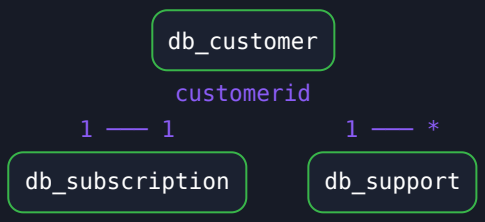
DB_SUBSCRIPTION
10,000 rows
Plans, charges & churn

DB_SUPPORT
6,494 rows
Complaints & CSAT

▸ Tables

TABLE	CONTENTS
db_customer	Customer demographic and profile information (ID, name, country, state, gender, date of birth).
db_subscription	Subscription plans, contracts, charges, churn and cancellation information (start/renewal/cancellation dates, plan, contract, monthly charges, CLTV, churn score).
db_support	Customer complaints, escalation status and CSAT information (complaint date, escalation flag, satisfaction score).

▸ Data Model



Three connected tables joined on **customerid**, with cardinalities matching the Power BI model: one customer to one subscription, one customer to many support records. The Excel workbook was first converted to a SQLite database, then queried into pandas.

▸ Business Logic — Defining Churn

Cancellation date present → Churned
A recorded cancellation date means the customer ended the subscription.

Cancellation date blank → Retained
No cancellation date means the customer is still active at the analysis date.

This rule was implemented as the engineered **churn_flag** column (1 = churned, 0 = retained) and drives every churn KPI in the project.

05 · METHOD

TECHNOLOGY STACK

Python
Core language for the entire analysis pipeline

Pandas
DataFrames, cleaning, joins & feature engineering

NumPy
Vectorized logic for flags, conditions & tenure

Matplotlib
Core charting for EDA

Seaborn
Statistical visuals, heatmaps & distributions

SQLite
Workbook → relational database conversion

Jupyter Notebook
Documented, reproducible analysis environment

Power BI
Data modeling & interactive dashboard

DAX
KPI measures: churn rate, ARPU, revenue at risk

Workflow

1 Data Ingestion → 2 Data Cleaning → 3 Feature Engineering → 4 Exploratory Analysis →
5 KPI Development → 6 Power BI Data Modeling → 7 Dashboard → 8 Insights & Recommendations

▸ Two-Tool Design

Python handled the heavy data work — restructuring the workbook into SQLite, cleaning inconsistencies, deriving features and running the EDA. Power BI then consumed the modeled data to compute DAX measures and present the KPIs and visuals interactively, with slicers for plan, contract, subscription type and state. The notebook's exported dataset fed the report, so both tools tell one consistent story.

06 · PREPARATION

DATA CLEANING & FEATURE ENGINEERING

▸ Data Cleaning

Data type inspection & correction
Dates stored as text (e.g., dob, subscription dates, complaint_date) were parsed to proper datetime types.

Date standardization
Mixed date formats handled with mixed-format parsing and coercion; invalid values checked afterwards (0 null DOBs after parsing).

Missing / null value checks
3 missing country values filled using a state→country mapping built from non-null rows.

Category consistency checks
Gender values standardized ("Men"→Male, "Women"→Female); a duplicated "Refferal/Referral" spelling was identified for review.

Data quality validation
Row counts verified at every merge (10,000 subscriptions preserved after joins); Latest support records were identified per customer for customer-level escalation analysis.

▸ Feature Engineering

Churn Flag
1 when a cancellation date exists, else 0 — the analytical backbone of the project.

Customer Age
Derived from date of birth to enable demographic views.

Tenure (days)
Subscription start → cancellation date for churned customers; start → today for active ones.

Churn Risk
Low / Medium / High bands from the dataset's churn score (≥70 = high) for segment analysis.

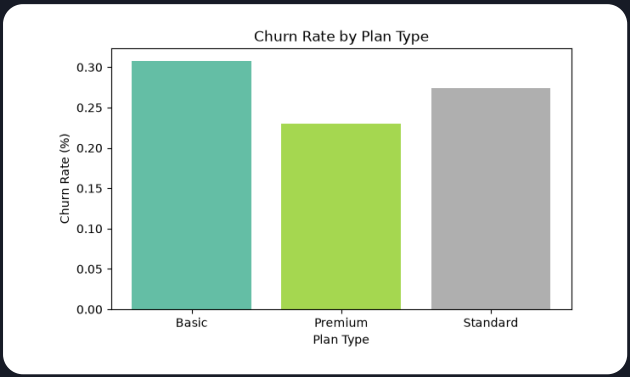
Churn Score
Used as provided in the dataset and validated against observed churn behavior.

Honest scope note: features here are rule-based derivations from existing columns and business logic. No predictive machine-learning model was trained in this project.

07 · ANALYSIS

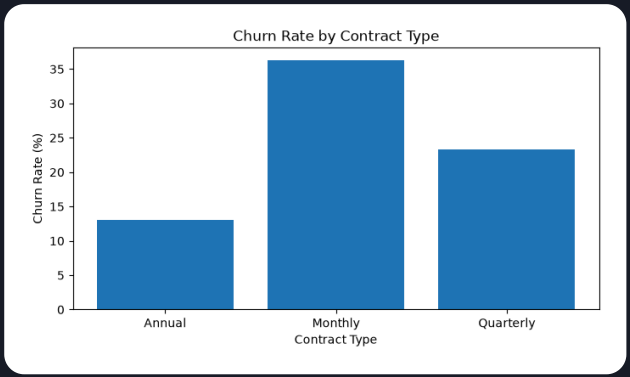
EXPLORATORY DATA ANALYSIS

Churn by Plan Type



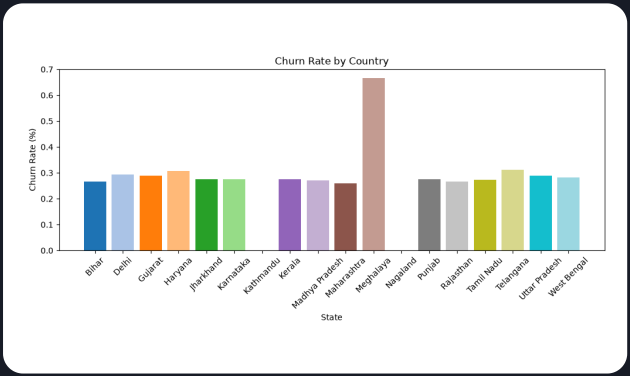
Basic plans churn highest (30.80%), followed by Standard (27.46%) and Premium (23.05%) — lower-priced plans show the most attrition.

Churn by Contract Type



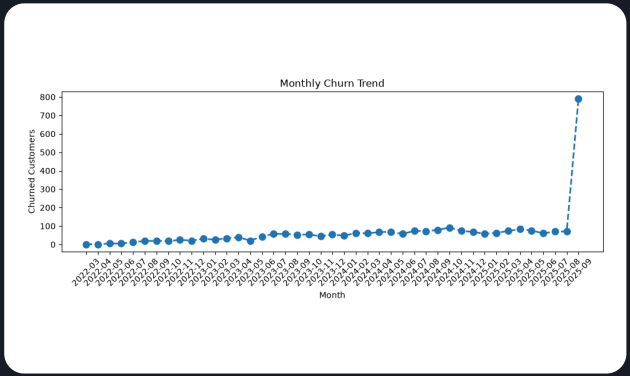
Monthly contracts churn far more than Annual ones; longer commitments are associated with materially lower churn.

Churn by State



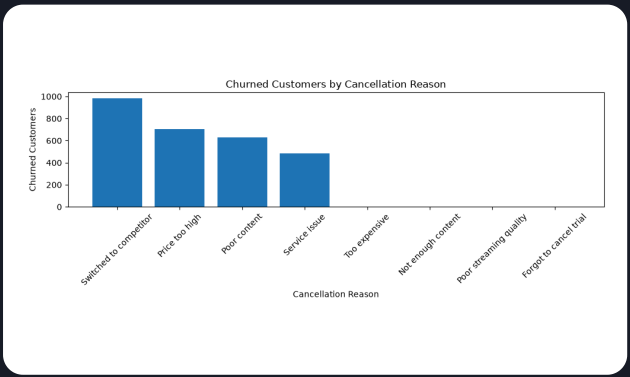
Churn varies across states — **Telangana (31.15%) and Haryana (30.78%) are highest**, Maharashtra lowest among large states (25.94%). Very small states show extreme but unreliable rates (n = 1–3).

Monthly Churn Trend



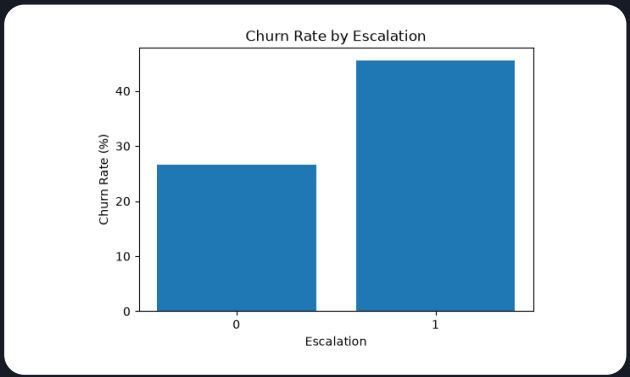
Cancellations are **not uniform over time** — a sharp spike appears late in the year, making the timeline worth monitoring.

Cancellation Reasons



"Switched to competitor" leads (986), then Price too high (703), Poor content (627) and Service issue (487). Long-tail reasons are negligible.

Support Escalation vs Churn



Churn is **≈ 64.98% vs ≈ 35.02%**. This is an observed association — escalation is a useful signal of dissatisfaction, not proven to cause churn.

Also examined in the notebook: a correlation heatmap across numeric fields, churn-score distributions, monthly-charge comparisons, CSAT analysis and gender-based churn analysis — all reflected on the dashboard.

08 · RESULTS

KEY PERFORMANCE INDICATORS

The headline metrics computed in the notebook and reproduced as DAX measures in Power BI.

CHURN RATE

28.07%

Share of customers with a cancellation date

RETENTION RATE

71.93%

1 − churn rate across all 10,000 customers

ARPU

12.96

Average monthly revenue per user

AVERAGE TENURE

≈ 934 days

Mean subscription lifetime (churned & active)

REVENUE AT RISK

35.05K

Monthly charges of churned customers

HIGH-RISK REVENUE AT RISK

15.35K

Monthly charges of customers with churn score > 70

ESCALATION RATE

7.69%

Share of customers whose latest support record was flagged as escalated

AVG COMPLAINTS / USER

0.65

Total complaints ÷ unique customers

▸ How the key KPIs were computed

Churn rate = mean of the churn flag × 100 · ARPU = total monthly charges ÷ unique customers · Revenue at risk = Σ monthly charges where churned · High-risk revenue at risk = Σ monthly charges where churn score > 70 · Escalation rate = share of customers whose latest support record was flagged as escalated · Avg complaints per user = total complaint count ÷ unique customers. The same definitions were implemented in DAX for the dashboard cards.

09 · DELIVERY

POWER BI DASHBOARD



▸ KPI Cards

Churn Rate · Retention Rate · ARPU · Average Tenure · Revenue at Risk · High-Risk Revenue · Escalation Rate · Avg Complaints/User

▸ Visual Analysis

Churn by Escalation Status · Churn by Churn Risk · Monthly Charges by Churn · CSAT by Churn · Churn by Gender · Cancellation Reasons · Monthly Churn Trend · Churn by State

▸ Filters

Plan Type · Contract Type · Subscription Type · State — every KPI and visual responds to the slicers for interactive exploration.

The screenshot above is the actual delivered dashboard — a single-page report in the same dark/green visual language used throughout this document. DAX measures power the KPI cards so figures stay consistent when filters are applied.

10 · INSIGHTS

KEY FINDINGS

▸ OVERALL CHURN

Overall churn is **28.07%**, while retention is **71.93%**. More than one in four customers in the dataset has cancelled.

▸ HIGH-RISK CUSTOMERS

Customers classified as **high churn risk (score ≥ 70) show substantially higher churn** than low/medium-risk groups — risk banding tracks real behavior.

▸ REVENUE AT RISK

35.05K in monthly charges is associated with churned customers; **15.35K** sits with currently high-risk customers who have not yet cancelled.

▸ CANCELLATION REASONS

Switched to a competitor is the largest cancellation reason (986), followed by pricing (703), content (627) and service-related (487) reasons.

▸ SUPPORT ESCALATION

Customers associated with support escalations show a substantially higher churn rate (**≈ 64.98% vs ≈ 35.02%**). This is an observed association and may indicate that escalation is a useful signal of customer dissatisfaction; it does not prove that escalation causes churn.

▸ GEOGRAPHIC & MONTHLY PATTERNS

Churn **varies across states** (Telangana & Haryana highest among large states) and shows **noticeable monthly variation**, including a late-year spike — both merit further investigation.

Reading the evidence carefully: findings 5 and 6 describe associations and patterns observed in the data. The project does not claim that escalations, plans or states *cause* churn — controlled analysis would be needed to establish causation.

BUSINESS RECOMMENDATIONS

Practical actions tied directly to the findings. Each recommendation points back to a specific pattern in the data.

1. PRIORITIZE HIGH-RISK CUSTOMERS

Use churn-risk indicators (score ≥ 70) to flag customers who may benefit from proactive retention efforts — outreach, offers or check-ins before cancellation.

2. ADDRESS COMPETITOR SWITCHING

Since switching to a competitor is the top reason, review pricing, content value, plan flexibility and retention offers for customers showing signs of switching.

3. IMPROVE SUPPORT RESOLUTION

Escalated cases are associated with much higher churn. Investigate escalated support cases and prioritize faster resolution and follow-up.

4. ADDRESS CONTENT & VALUE CONCERNS

Analyze content gaps behind "poor content" and "price too high" cancellations; consider personalized recommendations, flexible plans and targeted offers.

5. PROTECT REVENUE AT RISK

Prioritize retention efforts by combining churn risk with customer revenue/value — the 15.35K tied to high-risk customers is the most immediate exposure.

6. INVESTIGATE REGIONAL & SEASONAL PATTERNS

Dig into states and periods with elevated churn (including the late-year spike) before rolling out localized or timed interventions.

12 · REFLECTION

PROJECT OUTCOME

The project demonstrates an end-to-end analytics capability: taking raw, messy data through preparation, analysis, modeling and visualization all the way to business-ready insight.

Data Cleaning	**Feature Engineering**	**Exploratory Analysis**
KPI Development	**Data Modeling & DAX**	**Dashboard Development**
Business Insight Generation	**SQLite & SQL Queries**	**End-to-End Analysis**

Pipeline

RAW DATA Excel workbook → **PYTHON** Cleaning & analysis → **FEATURES** Flags, tenure, risk → **EDA** Patterns & KPIs →

POWER BI Model & dashboard → **BUSINESS INSIGHTS** Findings & actions

▸ What I Learned

Building this project developed practical, hands-on skills across the analytics workflow: writing clean pandas/SQL for real data quality issues, deriving meaningful features from business logic, designing KPIs that stakeholders actually read, modeling data and writing DAX measures in Power BI, and — most importantly — translating analytical results into business recommendations without overclaiming what the data shows. It also taught discipline: validating row counts after every join, standardizing categories, and separating association from causation in every insight.

13 · CLOSING

CONCLUSION & FUTURE SCOPE

▸ Conclusion

This project demonstrates how customer, subscription and support data can be transformed into actionable churn and retention insights. By combining Python-based analysis with Power BI visualization, the project identifies important churn patterns, high-risk segments, cancellation drivers and revenue exposure.

The resulting dashboard provides an interactive way to explore these patterns and supports data-driven thinking around customer retention.

Future Scope

1. Predictive churn modeling
2. Customer lifetime value analysis
3. More detailed customer segmentation
4. Automated retention workflows
5. Real-time monitoring
6. Deeper investigation of regional & seasonal churn drivers

Scope note: these are future possibilities only — they are not part of the delivered project and have not been implemented.

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